NTG | VIT VELLORE · AIRTEL SYNC 2026

UdyamOne · Solution architecture

Main components and data flows of the secure pilot foundation

Implemented local flow
Optional network / account path
Future integration

Public home + customer dashboard · web or mobile

Needs, reviewed tools, consent, orders and tracking
Phone OTP sessions; prerequisite-gated customer journey
Four-language advisor; detailed planning remains English

Role-filtered staff workspace

OIDC roles gate KYC, activation, billing and analytics
Partner sees only signed referral attribution
Operator evidence is omitted from customer responses

Optional online helpers

Network availability required

Microsoft TTS; device fallback
Optional Anthropic explanation on legacy quotes only
Quote facts only; output guarded
V3 plan text is deterministic.

needs / consent / orders
plan, findings and status

evidence / pilot outcomes
order and assistance views

PILOT DEPLOYMENT · ONE FASTAPI SERVICE + POSTGRESQL

API + access checks

app.py · validates requests; checks OTP / OIDC roles; routes every module below

HTTP / JSON, multipart uploads, static HTML / JS / CSS and provider webhooks

profile +
reviewed needs

Failure → re-plan → fresh owner acceptance

consented uploads / returned findings

provider requests / validated callbacks

Needs planner

planner.py + engine.py

Keep / Add / Replace Later
Eligibility and dependencies
Compare 12-month new costs
Reads product catalogue CSV

plan

Quote + order

app.py

Recompute plan + fingerprint
Require owner acceptance
Create order + private token
Seed service activation tasks

seed tasks

Activation + recovery

activation.py + app.py

Service / site dependencies
Versioned, deduplicated events
Keep independent progress
Failures trigger a new quote

blockers

Assistance + pilot

assistance.py + app.py

5,000 leads → RF propensity
Blocker → skill / channel
Assumed effects + pilot data
No automatic staff dispatch

orders +
accepted plans

tasks / events

pilot records / costs

Bill OCR + Smart KYC

kyc.py + planner.bill_draft

Bill fields → owner review
KYC verdict → bound order
Mock GST / Udyam CSVs
Temporary images deleted
KYC consent events logged

verdict

PostgreSQL · shared persisted state

Alembic migrations · local SQLite compatibility for demos

Orders, accepted plans, customer sessions and OIDC bindings
Activation tasks / events and plan revision history
Consent, registry/payment evidence, WhatsApp and pilot records

state

Provider adapters

providers.py / integrations.py

HTTP / auth / scoped evidence
Signed callbacks + dedupe
WhatsApp: consent + plans
No chat orders or payments

calls /
webhooks

Provider accounts pending

Adapters coded; live tests pending

Cashfree: registry evidence
Razorpay: TEST payment links and signed payment callbacks
Meta: signed WhatsApp messages and replies; shared planner
No live provider success claimed.

consent

Consent events

PostgreSQL audit records
Purpose, notice, outcome and codes
No images or registration IDs

Controls that protect the decision

Fresh acceptance for recovery · uploads removed after OCR · OTP and OIDC role checks
Checklist progress is not live activation; pilot assignment does not contact customers.

Future: Airtel provisioning

Approved API contract, product mapping and service checks needed.

Read left to right: review needs → accept a plan → track each service → route help.

Solid paths show implemented logical data flow within one service. Arrows do not imply separate microservices or live telecom activation.